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MODULE WISE ARCHITECTURE OF GEMS PROJECT 2.0

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GEMS 2.0 PROJECT

About the project:

The aim of the project is to develop a fully-fledged ERP system to manage the entire activity of GEMS. This project includes the management of the flow of all the activities from head office to centres.

Proposed system:

As per the requirement we are planning to build a web application for the entire management of GEMS. The newly proposed system will be web based and according to modern technology. In addition to the existing aspect, the system comprise of better user interface, better performance, more flexible royalty management module, mobile application for centres and students, certificate management and exam portal. We intend to construct the complete system in futuristic manner.

Overview

This document includes the module-wise architecture of GEMS PROJECT 2.0. On the go this document gives a high-level overview of the structure of the activities included in this project.

Modules

The below-mentioned are modules included in this project.

I. Head Office
II. RO
III. RSO
IV. ASO
V. Centre
VI. Certificates
VII. Exam portal
VIII. Mobile app
IX. Gobsbank
X. International Certifications
XI. Marketing
XII. Materials
XIII. Accounts
XIV. Inventory
XV. Ticketing

Head office:

The entire administration management of all the activities under all the departments of the organization will be handled by the HO.

Below mentioned will be activities handled from the head office.

User management

User type creation

Multiple user types can be created. Permissions can be assigned to the user type created. Based on the user type given, the user will inherit those permissions. We can assign a user type while creating a user.

User creation

Users can be created by sending the user invitation. User type will be assigned to the users and the user will inherit permission given in that user type. User can log in to their account by using the username and password received in the invitation mail.

Centre management:

Centre creation

A new centre can be created by the admin. A username and password will be generated for the centre created. Admin will have the provision to activate and deactivate the centre. Only at the active state centres will be able to log in to their account. Centres will be functioning based on the class assigned. Fee for the courses of the centres can be varying based on this class.

Class:

Class is the criteria to set the fee amount for different courses based on the region centre are located.

Class creation

The class can be created by choosing the region and setting the amount of the course for that region.

Staff Management

Course management

Division

Divisions are the categories different courses belong to, where course will be the items under that category. For e.g.: multimedia can be the category and Photoshop (course) will be item under that category multimedia.

Division creation

A new division can be created and while creating a course these divisions can be mapped to it.

Course:

Course creation

A new course can be created by giving the course name, course code, division, class, Fee details etc. Based on the class mapped to the course the fee can be varying for the centres.

Course Search

Created courses can be searched based on different criteria such as course name, course code, division, class etc.(new criteria can be added).

Scheme:

Schemes are created for giving different offer criteria’s. Royalty amount to be deducted can be given in each scheme.

Scheme creation

New scheme can be created and managed. The below mentioned scheme can be provided. Discount can be given either to one course or multiple courses.

I. Normal (full payment has to be made)
II. Constant percentage (Percentage of discount given)
III. Constant amount (constant amount discount given for courses)
IV. Variable percentage (discount given based on the exams and other activities conducted by the centre)
V. Variable amount (discount given at the time of admission)
VI. **Scheme creation enable for multiple courses.**

Region (centre, district, state, country) can be mapped while creating a scheme and the scheme will only be applicable to the centres of that particular region. Scheme can only be applicable based on the date validity given.

Fee Management

Fee creation

Fee amount can be created for the courses from the head office and centres can view this. Based on the class given the fee amount for the course will be varying for the centres.

Fee transfer

Fee transfer from one student to another student can be managed in different scenarios like wrong fee receipt created etc.

Fee/Receipt Cancellation.

Fee Refund.

Miscellaneous Fee- Exam Fee, Material Amount, Voucher Cost, Fine etc…

Approvals

The head office users have different user types and user permissions. Based on the permissions assigned to the users they can give approvals that satisfy different eligibility criteria for centres and students.

Feedback (By the students via App)

Head office will collect the feedback form students and centres based on different criteria.

I. Centre feedback

Feedback given by different centres for various activities can be viewed.

II. Student feedback

Feedback given by students from different centres can be viewed.

Student search

Head office can search and view the list of the students of different centres and based on this list they can calculate further marketing activities.

Reports

The head office will have the provision to view various analytical report data of all the centres. The below mentioned are the reports that can be viewed.

Admission wise report

Reports of the division-wise admission details of students from different centres can be viewed.

Fee wise report

Fee wise reports of students from different centres can be viewed.

Student dropout reports

Reports of the students who have dropped out their course can be viewed.

Student lagging reports

Report of the students who have lagged their course can be viewed through HO login.

- Enquiry
- Admission
  - Scheme-wise
  - Date filters (Day /Week / Month / Year/ Period)
  - Course-wise
  - Division-wise
  - Centre-wise
  - ASO-wise
  - RSO-wise
  - RO-wise
  - State-wise
  - Country-wise
- Collection
  - Scheme-wise
  - Date filters (Day /Week / Month / Year/ Period)
  - Course-wise
  - Division-wise
  - Centre-wise
  - ASO-wise
  - RSO-wise
  - RO-wise
  - State-wise
  - Country-wise
- Down-Payment
  - Zero down payment
  - Amount-wise
- Fee Completed Students Report
- Duration Completed Students Report
- Drop-out report

- Leave report
- Re-registration report
- Centre transfer report
- Course Lagging Report
- Date Reports
    - Day
    - Week
    - Month
    - Year
    - Period
    - Centre-wise
    - ASO-wise
    - RSO-wise
    - RO-wise
    - State-wise
    - Country-wise
    - Fee Balance Report
    - **Miscellaneous Report**

Toppers List

Admission wise

Collection

Billing wise

Scheme wise

ASO/RO/RSO/State/Country wise.

Offers

Offers /discounts can be given to the centres. The below-mentioned offers can be included. Offers given to the centres can be dynamically managed through the HO login.

Inaugural offer

For the new centres Inaugural offers can be given for the first month for all the courses .

Special offer

Special discounts/ offers can be given when students take admission as groups or individuals.

Group discount

When bulk admission are taken from colleges Group discounts/tie-up offers can be given.

Offers for Multiple courses as well

Certificates:

Course completion certificates will be provided to the students if they have completed the course. Centres will send the request to dispatch the certificates and after all the eligibility checking process the certificates will be approved and dispatched by the head office to the centres.

Certificate Search

Certificates can be searched based on the different criteria such as course, date and register number. (New criteria can be added)

Certificate printing

Certificates can be printed and dispatched to the centres from the head office and if all the eligibility criteria are satisfied by the centre and the students.

Certificate dispatching

Centre place the request for certificate dispatching. If all the eligibility is satisfied the head office dispatches the certificates and the corresponding dispatching status will be available in the centre login.

- Certification
    - Apply CDF from Centre
    - Certificate Corrections
        - Course Name
        - Student Name
        - Course Duration
        - Other Corrections
        - Certificate dispatch status (Notification to be sent to Centre)
        - Certificate acknowledgement by Centre
        - Students to be updated on certificate receipt at Centre
        - Certificate issued (to student) status date

Certification Reports:

Certification Issued.

Certification Cancelled.

Certification Correction

Duplicate Certificate.

Certificate Lost.

- Scheme-wise
- Date filters (Day /Week / Month / Year/ Period)
- Course-wise
- Division-wise
- Centre-wise
- ASO-wise
- RSO-wise
- RO-wise
- State-wise
- Country-wise

Centre:

Below mentioned will be activities handled from the centre.

Login Notification alert to be delivered to the Centre Director/Assigned Official Contact Number.

- Enquiry
- Admission (convert Enquiry to Admission)
    - Admission ID or Student ID (for each student)
    - Photo Upload based on Admission ID
    - Registration Number based on Course
        - Student Management
            - To change all fields
        - Welcome Kit
        - Academic Kit
        - Scheme
            - Scheme Change
            - Scheme Correction (%)
        - Course Change
            - Upgradation
            - Degradation
        - Mode
            - Normal
            - Fast track
            - Super Fast track
        - Re-registration
        - Centre transfer
        - Drop-out
        - Leave Request
        - Clearance Certificate (CC) Issue
        - **Admission/Fee payment notification vis SMN/WhatsApp.**

Enquiry

Student details who have enquired about the courses provided by G-TEC can be entered as an enquiry. These enquiries generated can be converted in to admission.

Create enquiry

Enquiries will be generated for the courses provided by G-TEC. These can be added to the system as a new enquiry.

Search enquiry

Enquiries generated can be searched with different criteria such as name, Course and date (New criteria can be added). Also, different filters can be used to search the enquiry.

Admission

Admission details of the students for courses provided by G-TEC can be added. Admission can be managed in two ways.

I. Enquiries converted to admission

Enquiries will be already created in the system and from there this enquiry can be converted to admission.

II. Direct admission

Student will be directly added to the system as admission. When admission is created this way simultaneously the same student will be created as an enquiry.

Admission ID or Student ID

Student Management

Student search

I. Students can be searched based on different criteria's such as name, course, application number.(New criteria can be added).

II. List of students can also be obtained based on fee complete, course completed, certificate issued, fee not completed, exam conducted, cancelled, fee pending, batch allotted and batch not allotted. (New criteria can be added)

Student management

The below-mentioned student management activities can be done through the centre login.

Student actions

Student actions can be used to manage different criteria such as course completion, certificate issued, exam conducted and drop out.

If the student has completed the course then, the date of completion can be updated.

If the student certificate is issued then, the date of certificate issued can be updated.

If the student exam is conducted the, date of exam conducted on can be updated.

If the student is dropping out the course, we can update based on two criteria i.e., permanent dropout/temporary dropout.

Student centre transfer

If the student has to change the centre they are currently in then, the centre can send a student centre transfer request for to the head office.

Student drop out

If the student has decided to dropout the course they are currently in then, drop out details can be added based on two criteria, i.e., permanent dropout or temporary dropout

Staff Details

Certificates

Certificate can be issued to the students if all the criteria are satisfied by the centre and as well as the students.

New certificate

Centre can send request for the certificates after the eligibility checking for centre and student. If any of the eligibility is not satisfied then centre will not be able to send the certificate request to the head office. (eligibilities can be discussed).

Certificate status

If the centre has requested for the certificates to the head office then it goes through various verification process. The status of the verification process can be viewed by the centre.

Certificate dispatching

If all the eligibility criteria are satisfied, the head office dispatches the certificates to the centre. Centre can view the dispatch report the certificates by selecting a date range.

Reports

Based on the different activities handled by centre and head office different reports can be generated. Different reports based on enquiry, admission, royalty, certificates etc. can be viewed. The reports can be used for verification purpose.

Royalty amount

Royalty amount will be collected by head office from the centres for each admission. Once a student makes the payment to the centre for the course they have taken admission the fee amount will be split to the head office and the centre. The royalty amount will be transferred to the HO login and fee amount will be transferred to the centre account

Mobile app

A mobile app will be done for the centre/head office and students. The mobile app will provide a quick view of all the analytical reports for the head office users and students will be able to view their entire course and fee related information.

HO mobile App

A mobile will be done for the head office to view the analytical report's data of the activities of G-TEC. Each HO user will be having a separate login and based on the permission assigned to the user type , the users will be able to view the reports. Reports include the fee details, collection details, admission details etc.

Student Mobile App

A mobile app will be done for the students to view the details such as course chosen, attendance, fee status, syllabus covered etc. student will be exactly able to know the fee payment status and also students will be able to make the fee payment online through their account. If the certificate is issued for the students they can also download and view it.

Academics & Examinations

- New Courses
    - Module-wise courses
- Fee Structure Upload (Connection with Operation)
- Exam Schedules and Training Schedules
- Examination Notifications, Training notifications
- API Connection – to transfer the examination registration on Operation approval into the Examination Portal
- Student Data to be Uploaded to the Gobsbank portal (API) on approvals/ verification
- Provision to manage the Mandatory Certification or Vouchers against each Course

- Manage G-TEC Certified Trainer (GCT) IDs with Trainer Verification Approval option
- Option to add the trainers list as Bulk Upload
- Other International Examinations like IAB, ABMA Examinations and redirecting to the exam portals
- Option to export and import reports
- Option to upload results (or fetch results from the Exam portal (through API))
- Reports and filters based on
    - Courses
    - Type of Courses
    - Duration
    - GCT
    - Results
    - Exams etc.

International Certifications

- Vendor Management
- Inventory/Stock / Voucher/ License Management
    - To manage the stock of vouchers/licenses/products
    - Reports that can filter stocks based on
        - Date (Day, Week, Month, Year, Period)
        - Certification
        - Count
        - Validity etc
    - Stock availability status and count to be populated
    - Reminder/Notification on less stock availability
- Certification/ Project Management
    - Add/manage Certification
    - Mandatory Certification
    - Optional Certification
    - Additional Certification
- Voucher/ Licence Management
    - Voucher Order from Centres
    - Assigning voucher/license codes to centres (based on stock availability)
    - Examination/slot bookings for International Examinations
    - Managing Single/Bulk Import for Exam registration
    - Option to Upload exam results towards the Exams completed
- Reports
    - All type of reports to manage the entire inventory/stock/certification/exam reports

Materials (GMATS)

- Material/Product creation possible with code, price and stock details
- Division-wise stock management
- Search option
- Reports
    - Item-wise

    - Stock-wise
    - Centre-wise
    - ASO/RSO/RO-wise
    - State-wise
    - Country-wise
    - Non-purchase of material report
    - Analysis report
        - Daily, Monthly etc
    - Date-wise (Day, Week, Month, Year, Period, Financial Year)
    - Welcome Kit and Academic Kits to be ordered via Centres and to be assigned with Material Codes according to the quantity ordered.
- All request from Centres to be viewed in the HO login (as per the roles and permissions set)
- Centres should have option to view, search and order the materials according to the stock availability.
- Orders placed should have option to be viewed and printed
    - Orders once assigned should have option to update the status
- Online Payment integration to GMATS
- Courier and parcel service options can include in the order form. Can add the LR NO and no of bundles in Pod no in the dispatch details
- Option to update the dispatch status that can be viewed by the Centres
- On material receipt – status update to be viewed in Admin login

Accounts

- Facility to generate tax invoices showing particulars such as invoice number, date, party name, income details, hsn/sac details etc.
- Receipt transaction should be made on bill by bill basis.
- System must be able to generate party ledgers at any period with facility to identify overdue debtors.
- Types of Income
    - Registration Fee from Franchisee
    - Royalty from Franchisee
    - Other Income from Franchisee
    - EXAM FEE
        - IAB
        - DIFA
        - MFA
        - CAMP
        - IAB absentees Fee
        - NI Fee
        - Late submission Fee
        - Processing Fee
        - SAP
        - Supplementary fee
        - Improvement fee
        - Processing Fee
        - ONLINE EXAM FEE
        - DEMO

        - SEMESTER FEE
        - GCT FEE
        - TALLY
    - CERTIFICATION FEE
        - JAIN
        - PSC
    - CERTIFICATION CORRECTION
        - NORMAL
        - SAP
        - TALLY
    - DUPLICATE CERTIFICATE
    - ADDITIONAL CERTIFICATE
    - VOUCHERS
        - TALLY
        - MOS
        - ITS
        - QB
        - SAP
    - ADVERTISEMENT SHARE
    - IT MAGIC
    - IELTS
    - TRAINING FEE
    - OTHERS
- Reports
    - Should be able to generate various accounting reports such as day book, ledgers, trial balance, profit and loss account, balance sheet etc
    - Income, expense, assets and liabilities should be grouped as per schedule III of companies Act or we should be able to make such groups.
    - Trading profit and loss account and balance sheet should also be in such format.
- Expenses
    - There should be booking-approval-payment system with regard to expenses

Ticketing System

- Manage and resolve issues, requests, and workflows

Staff / HR Management

- Staff Details entry
- Staff Punching tracker
- Attendance
- Late In & Early Out
- On field Staff tracker using Geo Tagged Location.

Marketing

1. Performance Dashboard (Admissions, Collections, Billing & Staff Targets)

- Graphical representation of centre performance in terms of admissions, collections, and billings.

- Comparative analysis with previous month/year vs. current data.
- Include monthly target setting and performance index tracking for each staff member and centre.

2. Detailed Qualification Sub-Category

- Enable entry of specific qualifications like “BA English,” “BSc Mathematics,” etc., for better profiling.

3. Lead Source Trackings

- Include a field titled “How did you hear about G-TEC?” with options like Referral, Hoarding, Newspaper, Digital Media, etc.

4. Last Institution Name

- Capture the name of the student’s last studied school/college.

5. PIN Code Field

- Add a field to capture the student’s area PIN code for geographic analysis.

6. SMS / WhatsApp Communication Options

- Enable communication options for course updates, promotional offers, fee due reminders, and appointment alerts.

7. Course Bundling & Combo Offer Visibility

- Allow bundling of multiple courses under one admission.
- Display available combo offers or discounts based on selected course combinations.
- Display active offers in the student dashboard to promote multiple admissions.

8. Monthly Instalment Reminder

- Provide auto/manual fee reminder options via SMS or WhatsApp for installment-based payments.

9. Mobile-Compatible Reporting

- Ensure all dashboards and reports are optimized for viewing and usability on mobile devices.

10. Marketing Promo Dashboard

- Develop a dedicated dashboard for instantly sharing social media promotions to centres and directly to students.

11. Tele-Caller Module

- Include tele-caller daily reporting
- Follow-up record management with status updates
- Appointment scheduling with auto-reminders

12. Batch Suggestion & Allocation

- System-generated batch suggestions based on the selected course and preferred timing.
- Final allocation rights to remain with respective centres for flexibility.

13. Staff Attendance & Biometric Integration

- Integrate staff attendance with biometric systems to ensure accuracy and central tracking.

14. Social Media Engagement via Student Login

- Prompt students to follow official G-TEC social media platforms within their login panel to unlock specific features or content.